Chapter 2
Basic Measurement Concepts



Consistent and understandable terminology is essential in metrology. Uniform terminology facilitates comparison of results and uncertainties between metrology organizations and measurement practitioners. Despite the importance of correct terminology, misinterpretation of common terminology still abounds in scientific, engineering, and academic communities. This chapter covers basic measurement terminology and concepts that are necessary to proper understanding of measurement data and associated uncertainties. The difference between accuracy, precision, resolution, repeatability, reproducibility, uncertainty, and other terms are discussed. The International Vocabulary of Metrology (referred to as "the VIM"), fundamentally a dictionary of metrology terms, serves as the basis for common terminology.

2.1 Introduction

In this chapter we cover basic measurement concepts, including common terminology, types of measurements, the importance of uncertainty in measurement, and typical sources of uncertainty. The terminology and concepts provided here are not comprehensive, but references are provided for further reading. The material here, as well as the definitions outlined in Chap. 3, are intended to help provide a common language for practitioners of metrology.

2.2 Measurement Terminology

To properly analyze measurements and uncertainty, consistent terminology must be used. Prior to the first issuance of the GUM in 1993 (ISO/IEC Guide 98 1993), approaches to uncertainty analysis and the associated terminology were largely ambiguous (Kline 1985). Maintained by the same organization as the GUM, the

© Springer Nature Switzerland AG 2020

S. Crowder et al., *Introduction to Statistics in Metrology*,
https://doi.org/10.1007/978-3-030-53329-8_2

International Vocabulary of Metrology—Basic and General Concepts and Terms (JCGM 200 2012), known as the VIM, provides detailed definitions of many terms relevant to metrology and uncertainty analysis.

The VIM uses what it calls the "Uncertainty Approach" to measurement terminology. The "Error Approach" was the historical norm prior to publication of the VIM and was used to estimate a value as close as possible to a single true value. The Uncertainty Approach does not attempt to determine a single true value, rather, it assumes that information from the measurement permits an interval of reasonable values. The uncertainty approach is the preferred terminology for use by metrology organizations. Therefore, it is only natural that this textbook references definitions provided in the VIM. While the Error Approach terminology is sometimes still encountered in references such as American National Standards Institute (ANSI) and American Society of Mechanical Engineers (ASME) standards (ANSI/ASME PTC 19.1 1983; Abernathy et al. 1985), caution must be exercised when mixing older terminology with that prescribed by the VIM.

Basic terminology is reviewed in Sect. 2.2.1. General measurement terminology, along with terms from the uncertainty approach and the older error approach are given, followed by terms specific to calibration. For definitions of other terms not included here, the reader should refer to the VIM. Terminology presented here is not necessarily in alphabetical order; rather, frequently used terms are listed first, followed by less frequently used terminology. Verbatim definitions from the VIM are italicized, with additional clarification and description provided. Some definitions from the VIM have been merged for conciseness and readability.

2.2.1 General Measurement Terminology

2.2.1.1 Measurand

Quantity intended to be measured (VIM 2.3, p. 17).

The "measurand" refers to the physical quantity or object which is being measured.

2.2.1.2 True Value (True Value of a Quantity)

Quantity value consistent with the definition of a property of a phenomenon, body, or substance, where the property has a magnitude that can be expressed as a number and a reference (VIM 1.1, p. 2; VIM 2.11, p. 20).

The true value is the unique actual value or set of unique values representing the parameter of interest. Unfortunately, in practice, the true value is not known and cannot be determined because all measurements have uncertainty.

2.2.1.3 Measurement Accuracy

Closeness of agreement between a measured quantity value and a true quantity value of a measurand (VIM 2.13, p. 21).

"Accuracy" refers to the degree to which the result of a measurement conforms to the true value. Higher accuracy indicates a smaller difference between the measured value and the true value, whereas lower accuracy indicates a larger difference between the measured value and the true value. While the term "accuracy" is commonplace when speaking of measurements or equipment, it is only appropriate to use as a relative or qualitative term. Accuracy is never equivalent to resolution or precision. Because the true value of a measurand can never be known exactly, accuracy cannot be determined quantitatively.

2.2.1.4 Measurement Precision

Closeness of agreement between indications or measured quantity values obtained by replicate measurements on the same or similar objects under specified conditions (VIM 2.15, p. 22).

"Precision" refers to the degree of agreement between nominally identical, repeat measurements. Higher precision indicates repeat measurements are in better agreement (smaller spread between measurements), whereas lower precision indicates there is less agreement (larger spread between measurements). Precision says nothing about bias or offset but is a contributor to overall uncertainty. While use of the term is commonplace, it is only appropriate to use as a relative term.

Figure 2.1 illustrates the terms "accuracy" and "precision" using an analogy to target shooting. The bullseye represents the true value of the parameter of interest, and the points of impact represent the measured values. In the first frame, the shots

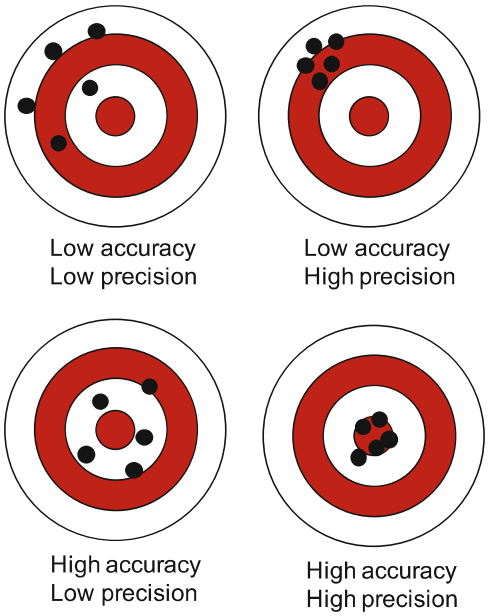


Fig. 2.1 Illustration of precision and accuracy

are found close together (more precise) but are far from the intended target (less accurate). In the second frame, the shots are found spread apart (less precise) but are centered around the intended target (more accurate). In the last frame, the shots are found close together (more precise) and are close to the intended target (more accurate). Note that the terms represent a qualitative, or relative, metric for describing the quality of a measurement.

2.2.1.5 Resolution

> *Smallest change in a quantity being measured that causes a perceptible change in the corresponding indication* (VIM 4.14, p. 42).

The resolution of an instrument represents the smallest change at which the measured value can be read or recorded. For an instrument with an analog readout, display, or scale, this represents the finest increment at which the measurand can be determined and depends on the spacing between consecutive markings and width of the indicator. For a digital readout or display, resolution represents the least significant digit to which the value can be read or recorded. In nearly every measurement discipline with any instrument, resolution never indicates overall accuracy or total uncertainty, but can be a contributor.

2.2.1.6 Measurement Repeatability

> *Measurement precision under a set of conditions that includes the same measurement procedure, same operators, same measuring system, same operating conditions, and same location, and replicate measurements on the same or similar objects over a short period of time* (VIM 2.20, p. 23; VIM 2.21, p. 24).

"Measurement repeatability" describes the degree of mutual agreement among a series of individual measurements, values, or results with the same operator, measuring system, and procedure over a short period of time (typically hours to days).

2.2.1.7 Measurement Reproducibility

> *Measurement precision under a set of conditions that includes different locations, operators, measuring systems, and replicate measurements on the same or similar objects* (VIM 2.24, p. 24; VIM 2.25, p. 25).

"Measurement reproducibility" describes the degree of mutual agreement among a series of individual measurements, values, or results with different operators, measuring systems, and/or procedures. Measurement reproducibility is often evaluated using measurements made over longer time periods. Measurement reproducibility and repeatability can be quantified using statistical techniques such as Analysis of Variance (see Chap. 10).

2.2.1.8 Independence of Measurements

The concept of "independence of measurements" is related to the concept of "statistical independence." To properly assess repeatability and reproducibility, a series of measurements (or observations) must not be related to one another. Typically this requires that repeat measurements are taken far enough apart in time that they are not related (or correlated in time). The goal is to allow only small random fluctuations on the measurement process. The conditions needed to achieve independence of repeat measurements are usually determined by subject matter experts in the measurement area. Section F.1.1 of the GUM (JCGM 100 2008) provides details regarding this concept.

2.2.2 Error Approach Terminology

The Error Approach to measurement uncertainty attempts to estimate the measurand as closely as possible to the single true value. Because the true value can never be known, the magnitude of the error can never be calculated exactly, and as such, use of the Error Approach terminology is discouraged. These definitions are included here because they are still frequently encountered in the literature, and they highlight the issue of inconsistent terminology. The terminology is summarized in Fig. 2.2.

2.2.2.1 Measurement Error

Measured quantity value minus a reference quantity value used as a basis for comparison (VIM 2.16, p. 22, VIM 5.18, p. 53).

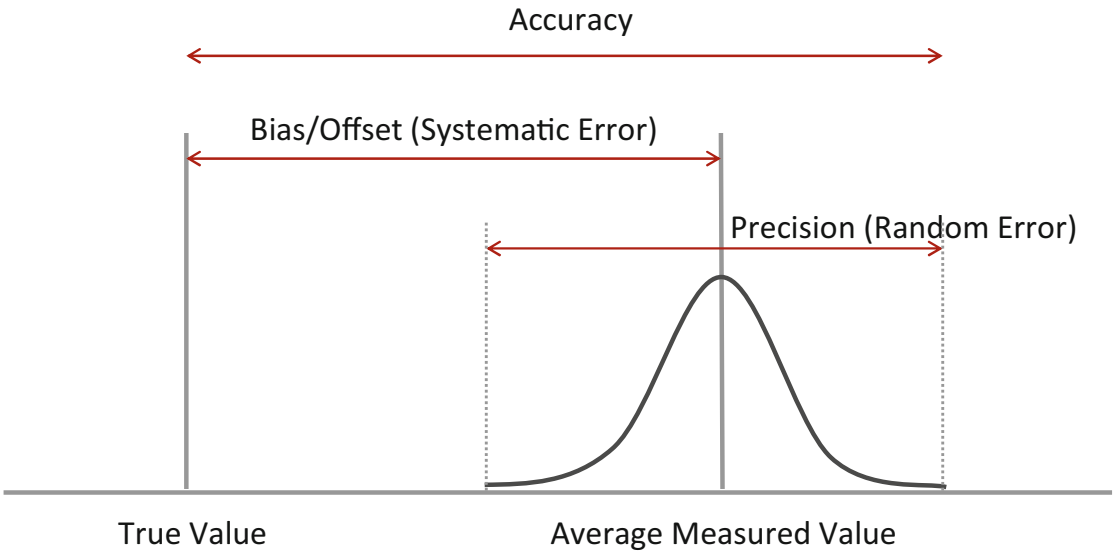


Fig. 2.2 Illustration summarizing terms for the older, discouraged "error approach" to measurement terminology

"Measurement error" is a term representing a quantitative value, indicating the magnitude difference between the measured value and the true value. However, since the true value can never be known, the exact error of a measurement can never be known. Therefore, use of the term "error," while commonplace, is not favored in the approach presented in the GUM or in this textbook.

2.2.2.2 Systematic Measurement Error

Component of measurement error that in replicate measurements remains constant or varies in a predictable manner (VIM 2.17, p. 22).

"Systematic error" is an error that, when fully quantified, is predictable. Systematic error can be a bias or offset in the measurement, or an error based on an independent parameter, such as a known temperature dependency. If the systematic error is known, a correction should be applied to the measurement result. Again, since the true value can never be known, use of the term "systematic error" is discouraged unless referring to a quantitative, known offset that can be corrected.

2.2.2.3 Random Measurement Error

Component of measurement error that in replicate measurements varies in an unpredictable manner (VIM 2.19, p. 23).

"Random error" is an error that, when fully quantified, is caused by factors that can vary from one measurement to another. Random error results from inherent variability in a measurement, process, or system, and cannot be corrected for since it derives from unpredictable variation in the measured value. Random error provides a quantitative measure of repeatability and reproducibility in a measurement and is typically determined stochastically. Use of the term "random error" is discouraged unless referring to a quantitative, known variability in a measurement system. Random error may result from noise or fundamental inherent variation from the measurement process, such as that observed when counting measurements of radioactive decay processes.

2.2.3 Uncertainty Approach Terminology

The Uncertainty Approach provides the preferred set of terminology for discussing measurement uncertainty. Instead of attempting to determine a "true" value, this approach assigns a range of likely values that could be attributed to the measurand based on the finite amount of knowledge of the measurement equipment and process.

2.2.3.1 Measurement Uncertainty

Non-negative parameter characterizing the dispersion of the quantity values being attributed to a measurand, based on the information used (VIM 2.26, p. 25).

Uncertainty means "doubt," and in a broad sense, the term "measurement uncertainty" means doubt in the validity of a measurement result. Since the true value can never be known, all measurements will possess uncertainty. Therefore, a measurement reported without an associated uncertainty is incomplete. Measurement uncertainty includes components arising from the uncertainties in the equipment and standards used to make the measurement, as well as from the measurement model and process itself.

Measurement uncertainty always includes Type A and Type B evaluations of measurement uncertainty. These concepts will be formally defined in Chap. 6. At a high level, Type A evaluations come from statistical analysis, and Type B evaluations are determined from other methods. Both Type A and Type B evaluations are required to determine the total measurement uncertainty. These terms refer to the method of evaluation and not the source of uncertainty associated with them, although it is commonplace to see uncertainties referred to as "Type A" or "Type B." These are not directly equivalent to the older terminology of "random error" and "systematic error," although in some instances they are used interchangeably.

Figure 2.3 depicts the relationships of measurement uncertainty. The true value is unknown, and therefore could fall anywhere within the estimated uncertainty, or even outside of the uncertainty estimate if the level of confidence is less than 100%.

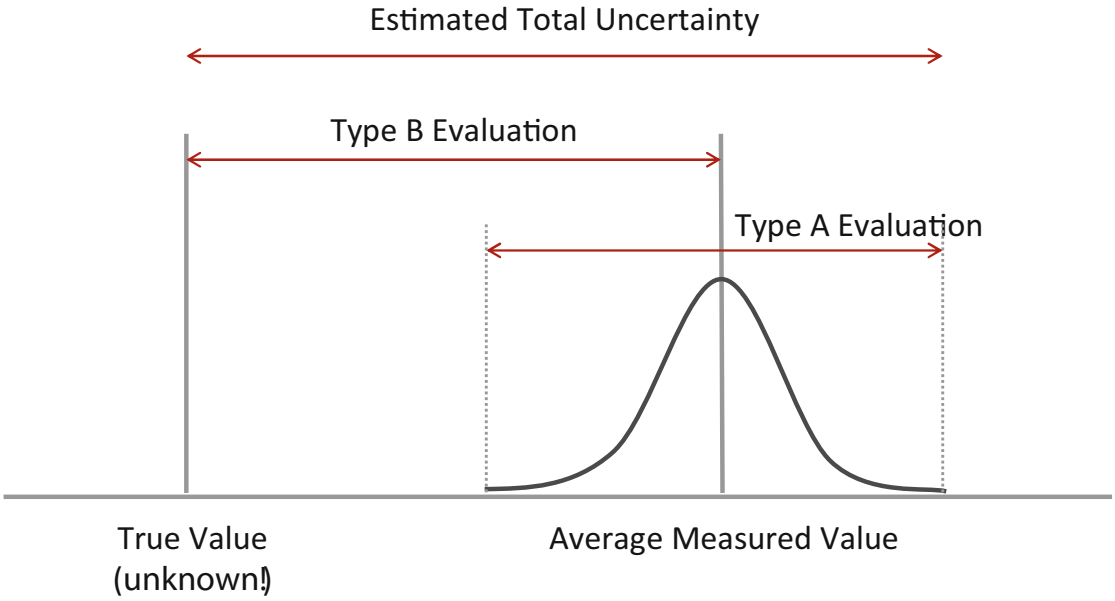


Fig. 2.3 Illustration summarizing terms for the preferred Uncertainty Approach to measurement terminology. The measured value will fall within the total uncertainty above or (not shown) below the true value

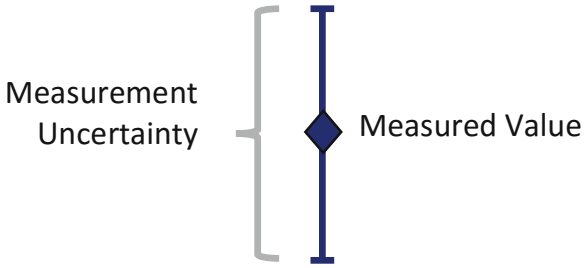


Fig. 2.4 Typical chart representation of measurement uncertainty using error bars

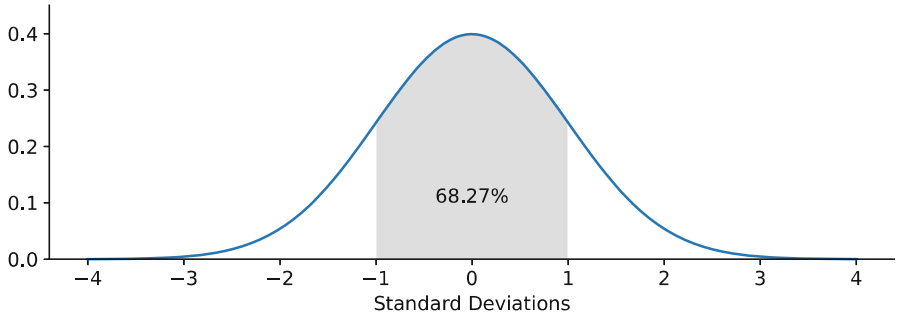


Fig. 2.5 One standard deviation under the normal curve (gray shaded region), corresponding to a level of confidence of 68.27% (often rounded to 68%)

Figure 2.4 depicts the common visualization of a measured value and its uncertainty using error bars.

2.2.3.2 Level of Confidence (Coverage Probability)

> *Probability that the set of true quantity values of a measurand is contained within a specified coverage interval* (VIM 2.37, p. 28).

In measurement terminology, the level of confidence is an important part of the measurement uncertainty. It represents the probability that the constructed uncertainty interval will contain the true value. The width of the uncertainty interval will change depending on the desired level of confidence, which should be reported along with the uncertainty interval.

The levels of confidence (coverage probabilities) for a normal random variable are often associated with multiples of the standard deviation, as illustrated in Figs. 2.5, 2.6, and 2.7.

2.2.3.3 Coverage Interval

> *Interval containing the set of true quantity values of a measurand with a stated probability, based on the information available* (VIM 2.36, p. 27).

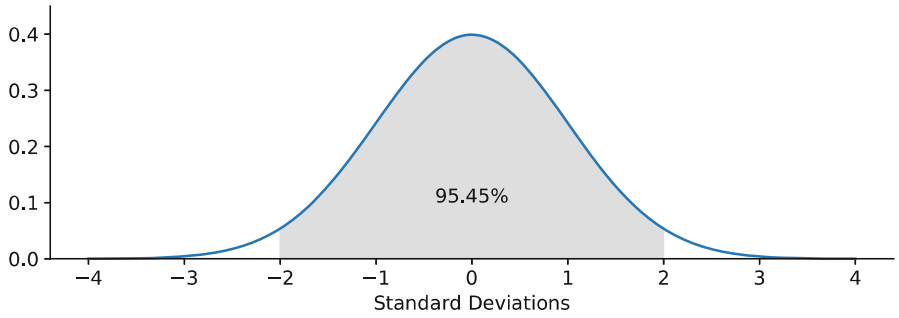


Fig. 2.6 Two standard deviations under the normal curve, corresponding to a level of confidence of 95.45% (often rounded to 95%). This is the typical value used on calibration certificates

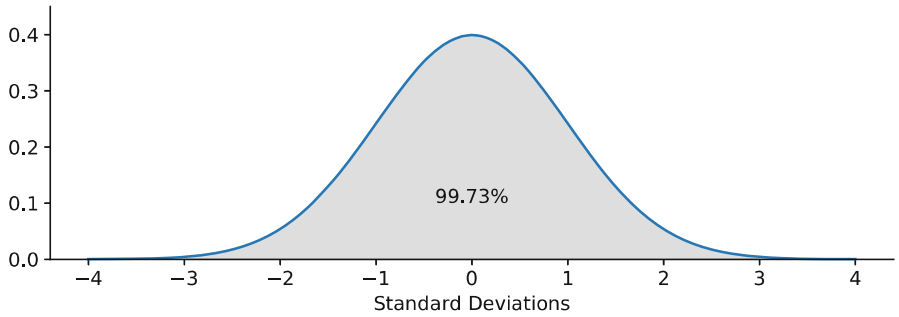


Fig. 2.7 Three standard deviations under the normal curve, corresponding to a level of confidence of 99.73% (often rounded to 99.7%)

The coverage interval is a range of possible values that includes the true value of the measurement with the specified level of confidence. The level of confidence is associated with the choice of coverage factor, discussed in later sections for both normal and non-normal distributions. Sometimes the term "confidence interval" is used; however, this term is not recommended, to avoid possible confusion with the statistical term of the same name.

2.2.3.4 Measurement Model

> *Mathematical relation among all quantities known to be involved in a measurement* (VIM 2.48, p. 32).

A measurement model is the formula used to combine measurements of different quantities in order to obtain the quantity of interest. Measurement models can be classified as direct or indirect (Chap. 6). A direct measurement refers to a measurement in which a single value is read directly from the measuring instrument (for example, a measurement of length using a ruler). An indirect measurement refers to a measurement obtained by measuring other quantities functionally related to the measurand. For example, when measuring electrical resistance, R, by making measurements of voltage, V, and current, I, the measurement model is Ohm's law

$R = V/I$. In this case, the value R is determined indirectly by first measuring two other quantities (V and I).

2.2.4 Terminology of Calibration

Relying on the uncertainty approach terminology described in Sect. 2.2.3, calibration has its own set of definitions.

2.2.4.1 Measuring and Test Equipment (M&TE)

Device or system to certify, test, or inspect items for conformance to requirements (DOE-STD-1054-96 1995).

2.2.4.2 Metrological Traceability

> *Property of a measurement result whereby the result can be related to a reference through a documented unbroken chain of calibrations, each contributing to the measurement uncertainty* (VIM 2.41, p. 29).

"Measurement traceability" refers to an unbroken chain of comparisons relating a device's measurements to a known standard. The top of the traceability chain should relate to one or more of the seven the SI base units. For details on traceability and the SI units, see Chap. 3.

2.2.4.3 Calibration

> *Operation that, under specified conditions, in a first step, establishes a relation between the quantity values with measurement uncertainties provided by measurement standards and corresponding indications with associated measurement uncertainties and, in a second step, uses this information to establish a relation for obtaining a measurement result from an indication* (VIM 2.39, p. 28).

Calibration is a periodic comparison of measurement values delivered by a device under test to a known standard or reference. The calibration results in a statement of the device's value, the associated uncertainty, and the level of confidence in that uncertainty. Although the calibration process is purely a comparison, the concept of measurement uncertainty is introduced by relating the accuracies of the device under test and the standard. Even though calibration and adjustments are often performed at the same time, calibration must not be confused with adjustment; an adjustment is a modification of the device's properties to ensure performance within specified limits. See Chap. 3 for more details on calibration.

2.2.4.4 Tolerance Test

A tolerance test is a pass or fail determination based on whether a measurement falls within required limits. A tolerance test may be considered a calibration if traceable reference standards are used in the measurement. See Chap. 5 for guidance on determining whether a particular measurement is adequate to perform a tolerance test.

2.2.4.5 Certification Uncertainty

The certification uncertainty is the uncertainty consisting of the measurement uncertainty and any uncertainties due to use, environment, handling, or variation with time along with the expiration criteria for the calibration.

Certification uncertainty is infrequently reported by commercial calibration laboratories; in fact, ISO/IEC 17025:2017 discourages reporting anything but uncertainty due to the measurement itself. However, certain industries require reporting a certification uncertainty and associated expiration date for all calibrations to appropriately account for uncertainty due to drift and handling between calibrations.

2.3 Types of Measurements

Measurements of a device can generally be divided into physical measurements and electrical measurements. A short overview of basic measurement types is provided here.

2.3.1 Physical Measurements

Physical measurements seek to estimate measurands such as dimension, mass, force, temperature, pressure, acceleration, material hardness, yield strength, fluid viscosity, thermal energy, and others. Such measurements may be direct or indirect (refer to Chap. 6). For example, using a Coordinate Measuring Machine (CMM) to measure the dimensions of a part constitutes a physical measurement (Fig. 2.8). Refer to Appendix C for a case study involving a CMM to determine the radius of curvature on a part. Another example of a physical measurement is the measurement of velocity using photometric methods.



Fig. 2.8 Metrologists setting up a dimensional measurement of a reference standard using a CMM

2.3.2 Electrical Measurements

Electrical measurements seek to estimate measurands such as voltage, current, resistance, capacitance, inductance, and others. Electrical parameters may be continuous (DC), pulsed, or time-varying (AC). The measurement of a high-current shunt is shown in Fig. 2.9. Voltage and current measurements using digital multifunction meters are commonplace in industry.

2.3.3 Other Types of Measurements

Chemical, radiation, and spectroscopic measurements all have associated uncertainty, but sometimes determining uncertainties is not straightforward. Some surface science measurements may be used to describe qualitative results (e.g., energy-dispersive x-ray spectroscopy [EDS] or X-ray photoelectron spectroscopy [XPS]) but can also be used to determine a quantitative result. Figure 2.10 illustrates a quantitative method called angle-resolved X-ray photoelectron spectroscopy, which is used to determine film thickness.



Fig. 2.9 Measurement of a high-current shunt

2.4 Sources of Uncertainty

All measurements, regardless of the type, have associated uncertainty. Primary contributors to measurement uncertainty include equipment, operators, environment, measurement models, etc. These types of measurement uncertainty will be highlighted here.

M&TE Equipment and instrumentation used to perform measurements is inherently inaccurate, regardless of whether it is calibrated or not. All equipment used to perform the measurement will contribute to the overall uncertainty. A calibration certificate seeks to quantify this component of uncertainty for the user of the equipment.

Standards and Calibration Calibrating M&TE against known reference standards, ultimately traceable back to the SI base units, is necessary to ensure equipment is functioning within its specifications, ensure equipment is traceable to known quantities, and establish uncertainties. Reference standards are not perfect, and every time a calibration is performed, the uncertainty of the reference standard becomes part of the uncertainty of the M&TE being calibrated. In general, the uncertainty of the M&TE can never be lower than the uncertainty of the standard

vary ~3% per °C, meaning that a leak rate measured 10 °C away from ambient could have a 30% offset. Piezoelectric transducers, such as those used to measure pressure or acceleration, have large temperature coefficients, and a correction (with associated uncertainty) must be applied to achieve reasonable measurements. Failure to apply such corrections results in gross measurement uncertainties. Electrical resistors drift with temperature, and many electrical instruments must be used within narrow operating ranges of humidity. M&TE typically has recommended shock and acceleration limitations that must be observed for proper function. Consistency in environmental conditions is critical and must be carefully considered.

Random Disturbances Measurements are subject to disturbances, which it may not be possible to eliminate, thus contributing to measurement uncertainty. For example, electrical instruments may be subject to electromagnetic interference, leading to undesired variability in the observed signal that is not attributable to the measurand of interest. All electronics generate some intrinsic electrical noise that cannot be eliminated even with proper shielding and grounding. Radiation counting experiments are subject to a "counting uncertainty" due to the probabilistic nature of radioactive decay and emission processes. Random disturbances may be modeled as additive noise which, when significant, must be accounted for in the uncertainty of the measurement.

Measurement Model Uncertainty In general, measurements make assumptions about the measurand being investigated. For instance, measurement of a linear dimension on a part may assume the part is perfectly straight, when it may not be perfectly straight. In this case, the model assumes perfection. More complex measurement models typically require more assumptions, leading to additional sources of uncertainty. For example, determining the velocity of an object by measuring the distance traversed over a certain duration assumes that the object was traveling at constant velocity. What if it were accelerating or decelerating? This would introduce additional uncertainty into the measurement due to the model assumption of constant velocity. Sometimes the model can be expanded to include additional factors, such as a temperature coefficient in a resistor measurement model, that eliminate some of the model uncertainty.

2.4.1 Evaluating Sources of Uncertainty

Ultimately, it is nearly impossible to determine all potential sources of uncertainty in a measurement process. However, the most significant uncertainty components can be captured by considering the sources above and using engineering judgement, experience, and published recommendations about a measurement.

2.5 Summary

Consistent, well-defined terminology is a critical component of metrology. This chapter has introduced common terms used when discussing measurement uncertainty to prevent confusion later in the book. While most readers are likely familiar with older terms used in the error approach, this book advocates usage of the uncertainty approach to terminology for compliance with modern standards and the GUM. It is important to also understand older terminology as it is still widely encountered in textbooks, journals, reports, and other documents.

Basic measurement concepts and an understanding of uncertainty sources are foundational elements for the practitioner of metrology. Understanding these concepts is also important for those who analyze measurement data, in order to realize inherent limitations of different techniques. Ultimately, experience performing a particular type of measurement provides the greatest understanding of uncertainty contributors to the measurement process. While this chapter does not delve into specific measurement disciplines, Appendix B presents related reading supporting achievement of valid measurements in a variety of disciplines.

2.6 Related Reading

The definitive reference for terminology used in metrology and uncertainty analysis is the International Vocabulary of Metrology, or VIM (JCGM 200 2012). Written in a dictionary-style format, it is an important reference but may not easily convey concepts to those without prior experience in metrology. Appendix D1 of NIST Technical Note 1297 (Taylor and Kuyatt 1994) provides further explanation and interpretation of the VIM terminology and serves as an important companion for those seeking a conceptual-level understanding of VIM terminology.

The direct and indirect measurement models are covered in greater detail in Chap. 6 of this book and in a draft GUM supplement (JCGM 103 2018) that is still in development. Also discussed is the concept of independence of measurements (or observations).

Kline (1985) provides a concise pre-VIM summary of terminology used in uncertainty analysis and explains important underlying concepts and the need for uncertainty analysis with any measurement. The paper also offers a number of examples and historical cases stressing the importance of uncertainty analysis in measurements.

An Introduction to Error Analysis: The Study of Uncertainties in Physical Measurements by Taylor (1997) provides coverage of important concepts and topics for those new to the field of metrology and uncertainty analysis. The Error Approach and Uncertainty Approach terminology is interspersed and used somewhat interchangeably throughout, as the first addition was written prior to the VIM. The text is especially useful for those with limited prior experience in statistics, data analysis, and uncertainty concepts.

John Mandel, a distinguished scientist and statistician at the National Bureau of Standards (renamed the National Institute of Standards and Technology in 1988), originally authored *The Statistical Analysis of Experimental Data* (Mandel 1984) in 1964. In addition to introducing key concepts such as measurement models and sources of variability, the book presents statistical concepts that are relevant to the experimentalist and metrologist, such as independence of measurement observations. Although utilizing the terminology of the Error Approach, the text serves as a good companion to this book.

While proper understanding of discipline-specific measurement processes and instrumentation selection are critical to obtaining good measurement data, these areas are not discussed at length in this book. Appendix B offers additional related reading for several metrology disciplines along with related reading for each area. For topical coverage of discipline-specific instrumentation and selection for the experimentalist, Holman (2012) provides an overview of various measurement principles and discussion of instrumentation options in the framework of reducing measurement uncertainty. With an emphasis on mechanical engineering, the text also provides coverage of basic electrical measurements utilized in engineering experiments.

Measurement uncertainty concepts discussed here are applicable to testing laboratories as well as chemical metrology. For additional information in these areas, see Kimothi (2002).

2.7 Exercises

1. Repeatability and reproducibility can both be used to assess measurement variability.

   a. Describe how repeatability differs from reproducibility.
   b. For the same type of measurement, which will represent larger variability, and why?
   c. Is reproducibility or repeatability a better representation of measurement variability? Justify your response.

2. A digital multimeter (DMM) with microvolt resolution reads to six decimal places, such that the measurement of a fully charged automotive battery (nominal 12.6 V) is 12.623172 V.

   a. Does the reading on the multimeter mean the measurement is accurate to a microvolt? Why or why not?
   b. What else could contribute to the accuracy of the automotive battery voltage measurement?

3. Consider the following scenario: You are at the pump filling your vehicle with gasoline. The gasoline is sold by volume at a price of $2.74 per gallon (1 US gallon = 3.785411784 L), as displayed on the pump. The total fill amount is 14.462 gallons as shown on the digital pump display.



Fig. 2.13 Emergency teams responding to the B-2 crash site

Investigation of the crash after the fact revealed that an error in calibrating certain Port Transducer Units (PTUs) resulted in a measurement bias of air pressure in excess of −0.40 in Hg (−1355 Pa).

Research the crash (both NASA and the U.S. Air Force offer documentation online) and discuss how the calibration error happened. Explain how such a small error in air pressure measurement led to the crash. What sources of uncertainty were not considered in the calibration of the PTUs?

References

Abernathy, R.B., Benedict, R.P., Dowdell, R.B.: ASME measurement uncertainty. J. Fluids Eng. **107**, 161–164 (1985)

ANSI/ASME PTC 19.1: Measurement uncertainty (1983)

DOE-STD-1054-96: Guideline to good practices for control and calibration of measuring and test equipment (M&TE) at DOE nuclear facilities (1995)

Forrest, E., Schulze, R., Liu, C., Dombrowski, D.: Influence of surface contamination on the wettability of heat transfer surfaces. Int. J. Heat Mass Transf. **91**, 311–317 (2015)

Holman, J.: Experimental Methods for Engineers, 8th edn. McGraw-Hill Companies, Inc., New York (2012)

ISO 1: Geometrical product specifications (GPS) — standard reference temperature for the specification of geometrical and dimensional properties (2016)

ISO/IEC Guide 98: Guide to the expression of uncertainty in measurement (1993)

JCGM 100: Evaluation of measurement data - guide to the expression of uncertainty in measurement (2008)

JCGM 103: Guide to the expression of uncertainty in measurement — developing and using measurement models (committee draft) (2018)